

# Purification of nucleic acids from agarose gels

## Risk assessment

- Transilluminators emit ultraviolet radiation, which can damage unprotected eyes and skin.
- Guanidine salts form TOXIC GASES with bleach or strong acid
- ▷ Wear gloves, safety glasses, lab coat
- DO NOT add bleach or acidic solutions directly into sample waste



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This is a bench card. Full protocol available online.

### >> Separation and excision of the desired DNA fragment from agarose gels

- (1.) Resolve the desired DNA fragment on a 0.6–1.2% agarose gel in TAE or lithium acetate buffer. Load at least 200 ng per lane. 

**Critical:** To avoid carry-over of DNA, use fresh running buffer and leave empty lanes between samples. 

**Critical:** Avoid TBE buffer if possible, since borate inhibits downstream enzymatic reactions. 

- (2.) Visualize the gel using a blue light or long-wave UV source in a darkroom or gel box.

**Critical:** Always use long-wave UV (> 400 nm) and minimize exposure. Avoid pre-imaging gels under UV to prevent DNA damage and pyrimidine dimer formation. 

- (3.) Excise the desired band with a clean surgical blade. Trim excess agarose to minimize the slice size. 
- (4.) Transfer the gel slice to a clean microcentrifuge tube.

### A > Fast flush extraction

- (1.) Place the gel slice into a spin column. Centrifuge at  $15\,000 \times g$  for 1 min. Collect the flow-through. 
- (2.) Rotate the column  $180^\circ$  and spin again at  $15\,000 \times g$  for 1 min.

**Quality assurance:** Use a blue light flashlight to check for residual fluorescence. If DNA remains, add 50  $\mu$ L 5 mM Tris pH 8.0 and repeat the spin. Expect a faint greenish glow in the eluate. 

- (3.) Combine all flow-throughs. Discard the column.
- (4.) *Optional:* Further purify by silica spin column or ethanol precipitation  SOP 0021. 

### B > Extraction with guanidinium thiocyanate

□  R0135 Buffer QG (Extraction buffer), 0.5 mL □ Isopropyl alcohol

- (1.) Weigh the gel slice.
- (2.) Add 350  $\mu$ L Buffer QG per 100 mg gel. If the gel slice weighs less, supplement with electrophoresis buffer to 100 mg.
- (3.) Melt the agarose at 40–60 °C. Mix occasionally. Cool to room temperature.
- (4.) If the phenol red indicator turns red, add 10  $\mu$ L 3 M sodium acetate pH 5.0.

**Critical:** Do not skip this step—binding will fail at neutral or basic pH. 

- (5.) Add 150  $\mu$ L isopropanol per 100 mg agarose. 
- (6.) Bind DNA to silica  SOP 0021. Skip Buffer PB. 

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🔗 [Recipe \(available online\)](#)   🛠️ [Troubleshooting \(available online\)](#)   📖 [Notes \(available online\)](#)

